Medicinal Chemistry I

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Learning objectives

- At the end of this session, you are able to:-
  - ✓ Define medicinal chemistry and its relationship with other disciplines
  - ✓ Discuss source and classification of drugs
  - ✓ Describe types of drug naming
  - ✓ Discuss physicochemical properties of drugs in relation to biological effect
  - ✓ Discuss the relationship between Stereochemistry, receptors and drug action
  - ✓ Discuss drug metabolism pathway

Prerequisites

- Organic Chemistry *
- Anatomy **
- Physiology **

* = Mandatory courses

**= Important courses

Introduction

- What is Medicinal Chemistry?
  - A bridge between chemistry and pharmacology
  - It is a chemistry based discipline, involving aspects of biological, medical and pharmaceutical sciences.
  - It is concerned with
    - The invention
    - Discovery
    - Design
    - Identification
    - Synthesis and development of new chemical entities suitable for therapeutic use

Cont'd

- The study of their metabolism
- The interpretation of their mode of action at the molecular level and
- The construction of structure-activity relationships.
- It also includes the study of existing drugs, their biological properties, and their quantitative structure activity relationships (IUPAC definition)

Generally Medicinal Chemists can:

- Make new compounds
- Determine their effect on biological processes.
- Alter the structure of the compound for optimum effect and minimum side effects.
- Study uptake, distribution, metabolism and excretion of drugs.

Introduction cont'd….

- What does a medicinal chemist do; professionally?
  - Drug designers
  - Metabolomics

Terms

- A drug is any substance or mixture of substances manufactured, sold or represented for use in diagnosis, treatment or prevention of a disease, abnormal physical state or the physical symptom thereof, in man or animal;
  - Restoring, correcting or modifying organic functions in man or animal
    - E.g. contraceptive pills

Terms cont'd…

- How about drugs in use by an addict on the streets?
  - Your ideas
  - "Drug addicts all over the streets of the city"
    - No real big difference for the medicinal chemist

Terms cont'd…

- Structure activity relationship (SAR)
  - ✓ Is the relationship between chemical structure and pharmacological activity for a series of compounds

Classification of drugs

- Drugs can be classified according to various criteria:

1. By origin—sources of drugs

- ✓ Drugs may be obtained from:- Plants,Animals, Minerals or Microorganisms.
- ✓ The drugs may also be semisynthetic or synthetic compounds.

- The sources of drugs are summarized as follows:

A. Synthetic:-Most of the drugs in use today are synthetic in origin.

- ✓ Such drugs are chemically pure and it is easy to maintain supply line.

Ex. : Aspirin, Paracetamol.

Classification…

B. Natural:- There are number of natural sources.

- ✓**Plants:-** A number of plant based drugs such as vincristine, taxol, digoxin, quinine, reserpine, ergotamine, ephedrine, colchicine etc.
  - Are still a part of standard therapy.
  - Most of these don't have any synthetic substitutes.
- ✓**Animal:-** Some modern drugs continue to be derived from animal sources because the synthesis of such chemicals is very cumbersome and expensive.
  - Ex. : Gonadotrophins, heparin, insulin, thyroid extracts and enzymes.

Classification…

C. Microorganisms:-

- ✓ Following the accidental discovery of penicillin from a mould in 1928 and its successful use in chemotherapy in 1940, a large number of antibiotics have been discovered from a variety of soil fungi and some bacteria.
- ✓ These drugs form the most important group of chemotherapeutic agents used against infective diseases.
  - ✓ Ex. : Penicillin, Streptomycin, Tetracycline.

D. Minerals:-

- ✓ Minerals of medicinal value are iron, calcium, magnesium, aluminium, sodium, potassium etc.

Classification…

C. Semi-synthetic:-

- ✓ In some cases, especially with complex molecules, the synthesis of a drug may be very difficult or expensive and uneconomical.
- ✓ At the same time, the ones derived from natural sources may be impure. In these cases semisynthetic processes are used.
  - ➢ Ex. : 6-Aminopenicillanic acid is obtained from the fungus Penicillium chrysogenum.

Classification of drugs

- By pharmacological effect
- By chemical structure
- By target system
- By site of action

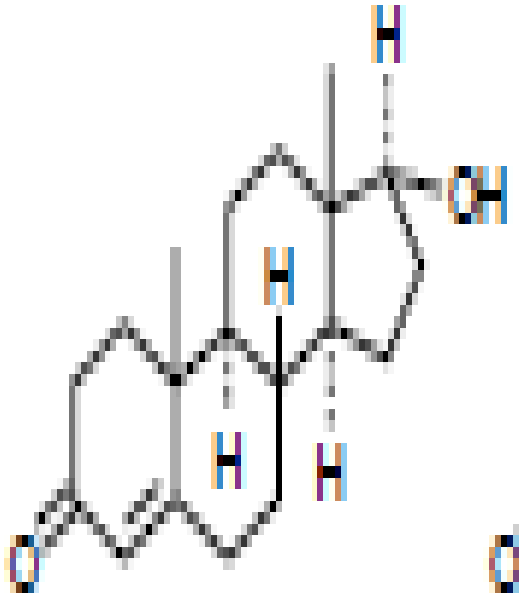
Pharmacological clasfn.

- According to their biological or pharmacological effect
  - Analgesics (pain killer)
  - Antipsychotics
  - Antibiotics
- Preferable, if you want to know the full scope of drugs available for a particular disease
  - Should not be confused as if all painkillers are the same because there is more than one way to target a disease

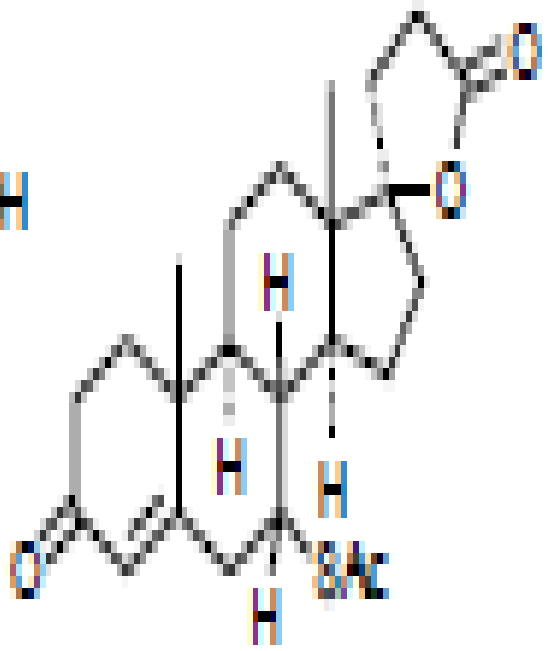
Chemical structure clasfn.

- According to the common skeleton drugs have
  - E.g. Penicillins, opiates, steroids
- Confusing;
  - In thinking that all chemical compounds of a certain chemical group have the same biological action

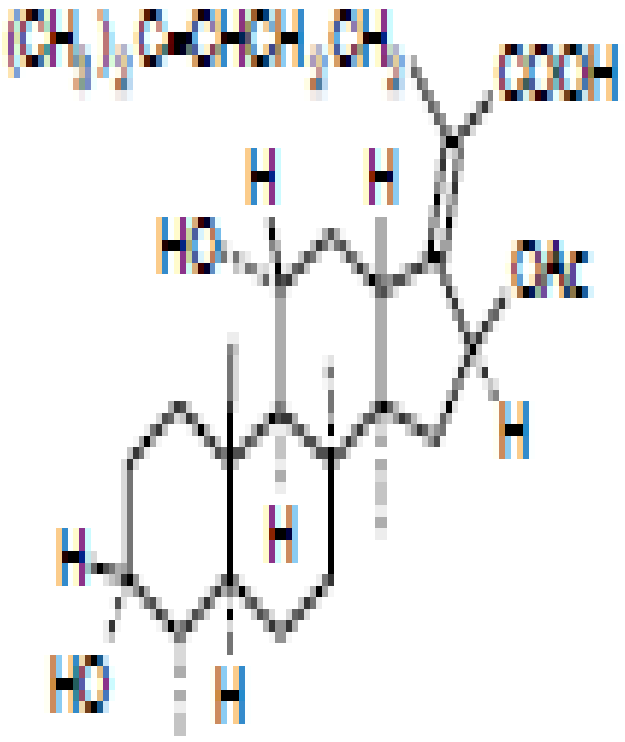
➢ E.g. Steroids, for example, have widely differing activities: testosterone is a sex hormone, spironolactone is a diuretic and fusidic acid is an antibacterial agent



Testosterone



Spironolactone



Fusidic acid

Chemical cont'd…

- Most drugs act at different sites of the body and have different physiological effect

- Useful to medicinal chemists who are concerned with synthesis and structure–activity relationships.

Target system clasfn.

- Affect a certain target system in the body, usually regarding neurotransmitters
  - E.g. antihistamines, cholinergics etc.

- A bit more specific than pharmacological classification

  - However, still a system with several stages to affect

Site of action classification

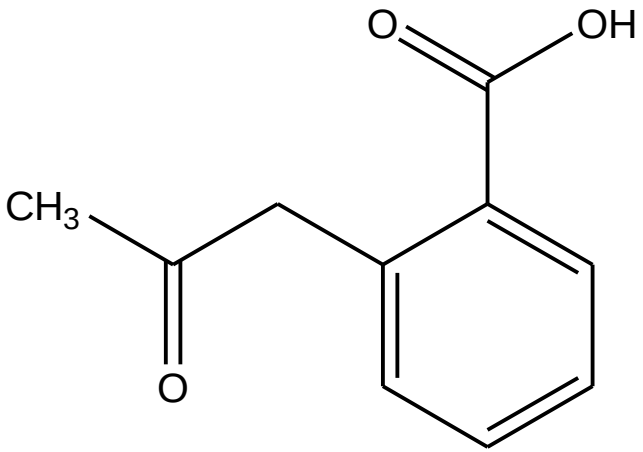
- According to the enzyme or receptor with which they interact
  - E.g. anticholinesterases
- More specific
  - Particular site the drug targets

Nomenclature of drugs

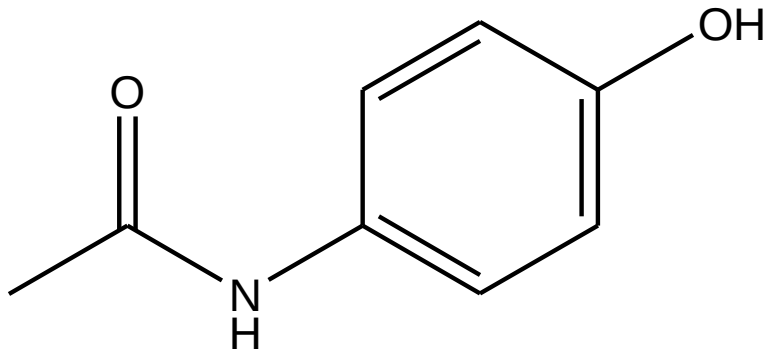
- Drugs can have a
  - Chemical name
  - Code number designations
  - Brand names
  - International non proprietary names

Chemical names

- Describes chemical structures
  - IUPAC naming
  - E.g. 2-acetoxy-benzoic acid, *N*-(4-hydroxyphenyl) acetamide



2-Acetoxy-benzoic acid



N-(4-hydroxyphenyl)acetamide

Chemical cont'd…

- Chemical name
  - Unambiguously defines a specific chemical substance
  - Often very long,
    - Difficult to memorize and
    - Practically incomprehensible
  - Gives no indication as to the therapeutic use

Code number designations

- Is usually used for experimental/ investigational drugs
- Is made with the initial letter of the name of the laboratory, the scientist or the medicinal chemistry research team that designed, synthesized and tested the drug followed by a number
- Supposed to be “covert naming”
  - E.g. MK-639-merck pharmaceutical,indinavir

Brand names

- Is a trade name
  - Are names selected by the owner of the product and registered in patent offices
    - E.g. coca cola, Addidas, Nike, “Bishan” Ambo, Yes Water, etc…
- They are private property and can only be used with the consent of the owner
  - Used for an active substance in a specific dosage form and formulation

International non proprietary names

- Common established name
  - As public property without restraint
- When drug patent is expired
- Generic drugs
- Designated by national or international commissions
  - Identifies the API as opposed to the finished product
  - E.g. omeprazole20
- The WHO provides and controls such names

non proprietary cont'd…

- Antihyperlipedemic agents
  - – *vastatin* for HMG Co-A reductase inhibitors
  - – E. g. Atorvastatin, simvastatin
- -*olol* for beta blockers
  - Propranolol
  - Atenolol

Physico-chemical properties in relation to biological action

- **Drug action** results from the interaction of drug molecules with either normal or abnormal physiological processes.
  - ✓ Drugs normally interact with targets (which they are proteins, enzymes, cell lipids, or pieces of DNA or RNA).

- The ability of a chemical compound to elicit a pharmacologic /therapeutic effect is related to:-
  - ✓ The influence of its various physical and chemical (physicochemical) properties

- The most pharmacologically influential physicochemical properties of organic medicinal agents (OMAs) are:
  - ✓ Solubility
  - ✓ Acidity and basicity
  - ✓ Reactivity

1- SOLUBILITY OF ORGANIC MEDICINAL AGENTS

❑ Importance of solubility:-

- ✓ Formulation of the drug in an appropriate dosage form and
- ✓ Bio-disposition: Disposition of OMAs in the living system after administration (absorption, distribution, metabolism, and excretion).

❑ The solubility expression: in terms of its affinity/philicity or repulsion/phobicity for either an aqueous (hydro) or lipid (lipo) solvent.

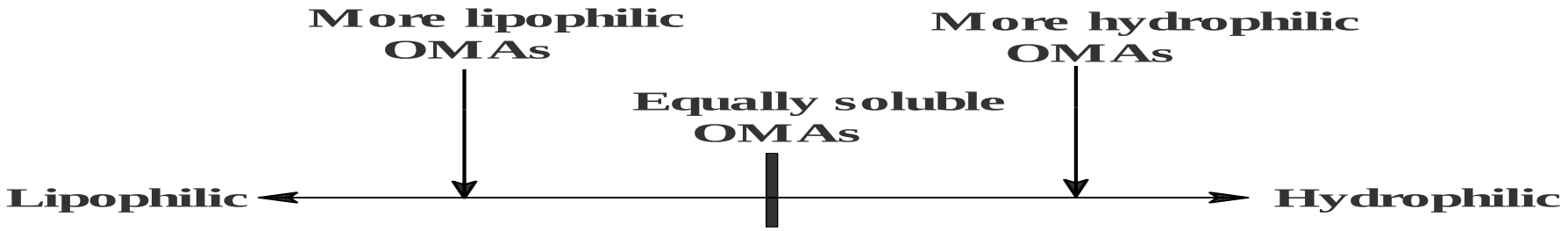
- ✓ Hydrophilic.....water loving
- ✓ Lipophobic.....lipid hating
- ✓ Lipophilic.....lipid loving
- ✓ Hydrophobic.....water hating

1- SOLUBILITY OF ORGANIC MEDICINAL AGENTS

- Majority of OMAs possess balanced solubility (have some degree of solubility in both aqueous and lipid media).
  - ✓ Because there is a need for OMAs to move through both aqueous (plasma, extracellular fluid, cytoplasm, etc.) and lipid media (biologic membranes) in the biological system.

SOLUBILITY OF ORGANIC MEDICINAL AGENTS

- Solubility of OMAs should be viewed as being on a continuum between high lipophilicity on one end of the spectrum and high hydrophilicity on the other.



SOLUBILITY OF ORGANIC MEDICINAL AGENTS

- In order for a chemical compound to dissolve in a particular solvent/medium the compound must establish
  - ✓ Attractive forces between itself and molecules of the solvent.

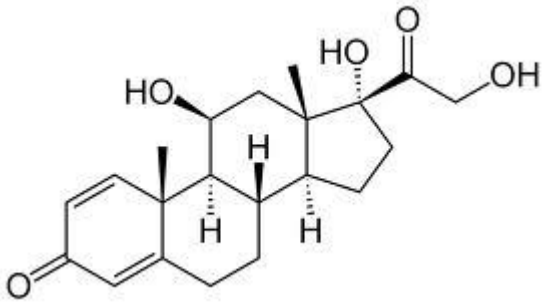
- It is possible to estimate the solubility properties of an OMA (hydrophilic vs. lipophilic) by examining the structure of the OMA and noting whether its structural features promote affinity for aqueous or lipid media.

The most important intermolecular attractive forces (bonds) that are involved in the solubilization process are:

1. Van der Waals Attraction

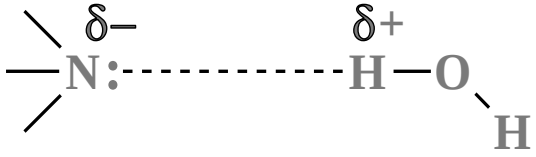
- ✓ Weakest intermolecular force (0.5-1.0 kcal/mole)
- ✓ Electrostatic
- ✓ Occurs between nonpolar groups (e.g. hydrocarbons)
- ✓ Highly distance and temperature dependent

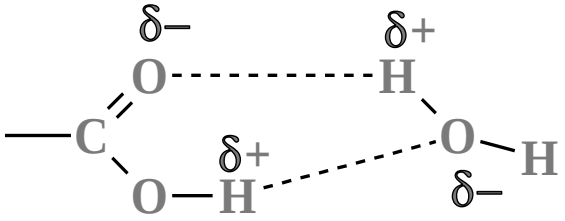
❖ **Is it really the weakest source of attraction?**



2. Dipole-Dipole Bonding

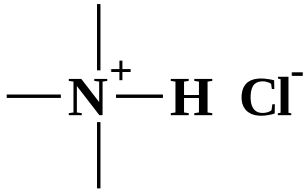
- ✓ Stronger (1.0 to 10 kcal/mole)
- ✓ Occurs electrostatically between electron deficient and electron excessive /rich atoms (dipoles)
- ✓ Hydrogen bonding is a specific example of this bonding and serves as a prime contributor to hydrophilicity

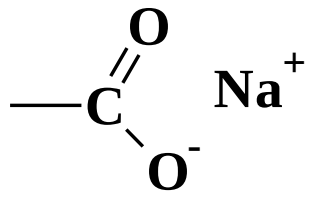




3.Ionic Bonding

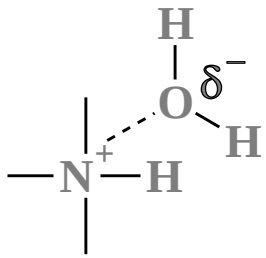
- ✓ Electrostatic attraction between cations and anions
- ✓ Common in inorganic compounds and salts of organic molecules
- ✓ Relatively strong (5 kcal/mole)

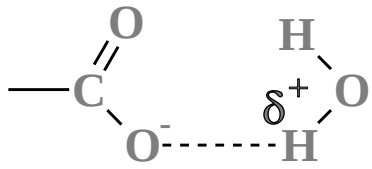




4.Ion-Dipole Bonding

- ✓ Electrostatic between a cation/anion and a dipole
- ✓ Relatively strong (1-5 kcal/mole)
- ✓ Low temperature and distance dependence
- ✓ Important attraction between OMAs and H2O





Solubility Prediction

- The relative solubility of an OMA is a function of the presence of both lipophilic and hydrophilic features within its structure,
  - ✓ Which serve to determine the extent of interaction of the OMA with lipid and/or aqueous phases.
- The relative solubility of an OMA can be determined in the laboratory,
  - ✓ **The partition coefficient**

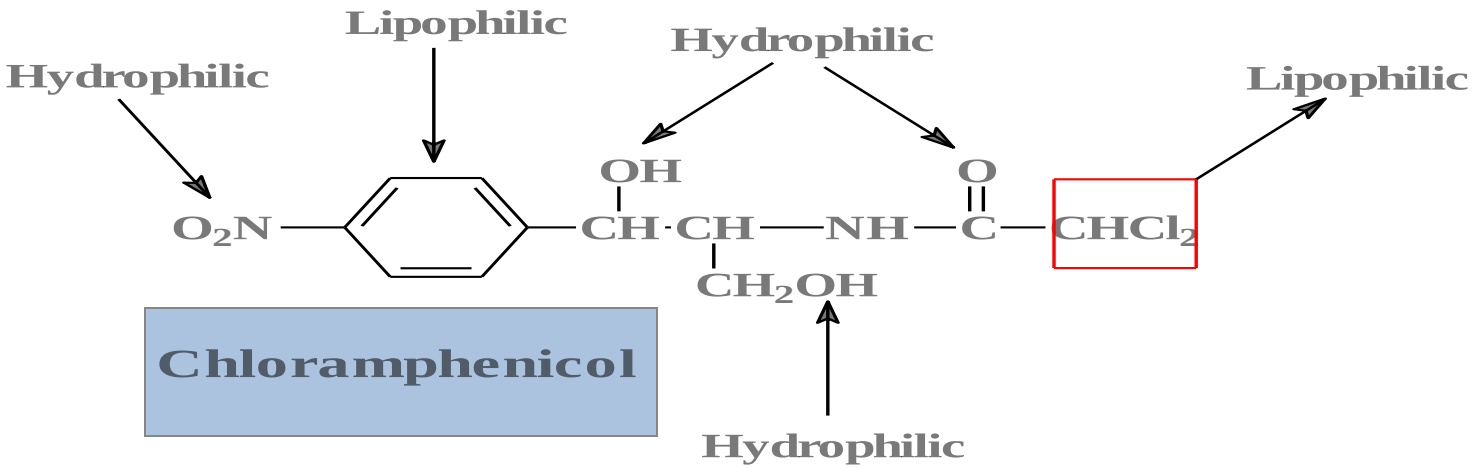
P=[Drug]lipid/ [Drug aqueous]

  - ✓ [**P**; the ratio of the solubility of the compound in an organic solvent to the solubility of the same compound in an aqueous environment
  - ✓ P is often expressed as a log value.

Solubility Prediction

- A mathematical procedures also have been developed to estimate the relative solubility of an organic molecule based upon differential contributions of various structural features to overall solubility
  - The relative solubility of an OMA is the sum of the contributions of each group and substituent to overall solubility.
- Example:
  - Examination of the structure of **chloramphenicol** (indicates the presence of both lipophilic (nonpolar) and hydrophilic (polar) groups and substituents.

Solubility Prediction



- The presence of oxygen and nitrogen containing functional groups usually enhances water solubility. While lipid solubility is enhanced by nonionizable hydrocarbon chains and ring systems.

Solubility Prediction

1. Laboratory Estimation of Relative Solubility

- The relative solubility of an organic compound is measured by determining the extent of its distribution into an aqueous solvent and a lipid solvent (usually n-octanol).
- These experiments generate a value, P, the partition coefficient for that particular compound.

$$\text{Partition coefficient} = \frac{\text{Conc. of compunds in } C_8H_{16}OH}{\text{Conc. of compunds in } H_2O}$$

2- Mathematical Estimation of Relative Solubility

- Solubility contributions (groups and substituents) are expressed as hydrophilic (negative value) or lipophilic (positive value) fragment constants.

$$\text{Log Pcalc} = \Sigma\pi$$

Where; Log Pcalc = log of partition cofficient and $\Sigma\pi$= sum of hydrophilic-lipophilic constants.

Hydrophilic-Lipophilic constants.

Fragments	π Value
C (aliphatic)	+0.5
Phenyl	+2.0
Cl	+0.5
O_2NO (nitrate ester)	+0.2
IMHB (intramolecular hydrogen bond)	+0.65
S	0.0
O == C–O (carboxyl)	–0.7
O == C–N (amide, imide)	–0.7
O (hydroxyl, phenol, ether)	–1.0
N (amine)	–1.0
O_2N (aliphatic)	–0.85
O_2N (aromatic)	–0.28

Calculation steps of Log *P* for OMA

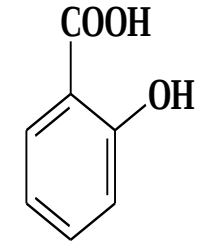
(i) The molecule is dissected into its various groups, functionalities and substitutents

(ii) Appropriate hydrophilic/lipophilic fragment constants are assigned and summed

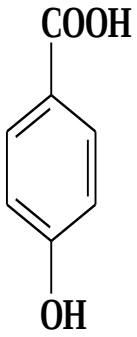
(iii) Compounds with **log *P* calc** values **greater than +0.5** are considered **water insoluble** (lipophilic) and those with **log *P* calc** values **less than +0.5** are considered **water soluble** (hydrophilic).

Calculated log P Values for salicylic acid and *p*-Hydroxybenzoic acid:

Salicylic acid		*p*-Hydroxybenzoic acid	
Fragment	**π Value**	**Fragment**	**π Value**
Phenyl	+2.0	Phenyl	+2.0
OH	-1.0	OH	-1.0
COOH	-0.7	COOH	-0.7
IMHB	+0.65	-	-
Sum	**+0.95**		**+0.3**
Prediction	**Water insoluble**	**Prediction**	**Water soluble**

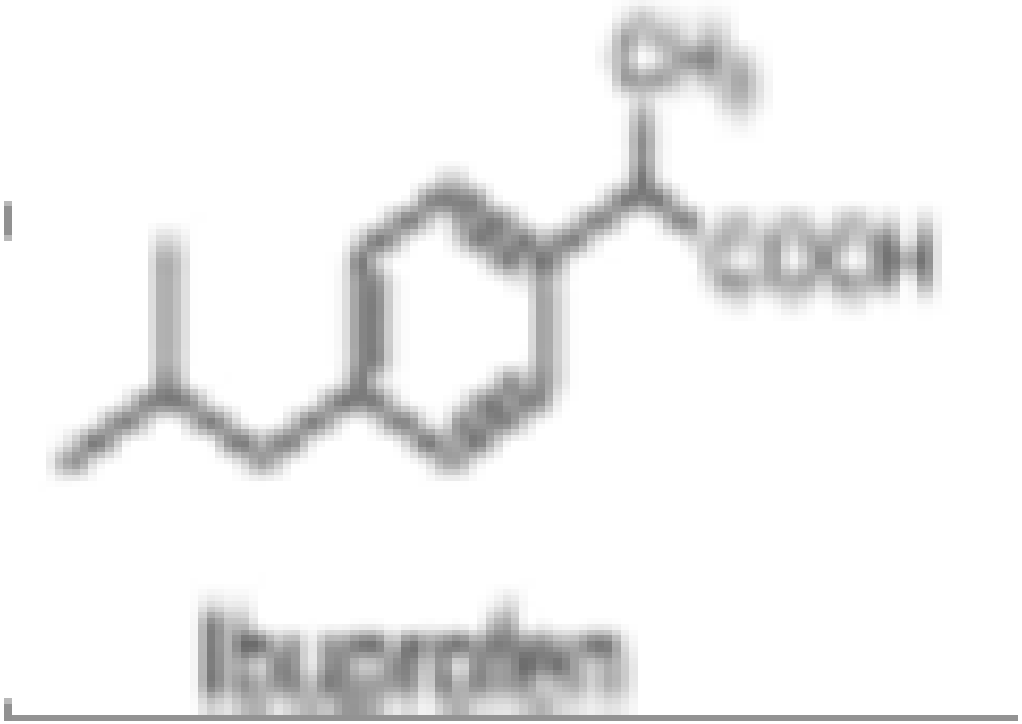


Salicylic acid



p-Hydroxybenzoic acid

Calculated log P Values for the following structure of ibuprofen



Fragments	π
6 carbons	+3.0
1 phenyl	+2.0
1 carboxyl	-0.7
logP	+4.3

2- Acidity and Basicity

- Acidic and/or basic properties of OMAs are important in both:
  - ✓ Pharmaceutical phase (dosage formulation, etc.) and
  - ✓ Pharmacological phases (disposition, structure at target site, etc.).
- The three aspects of acid-base chemistry:
  - ✓ Definitions
  - ✓ Recognition of acidic or basic organic functional groups and
  - ✓ An estimation of the relative acid/base strength of these groups

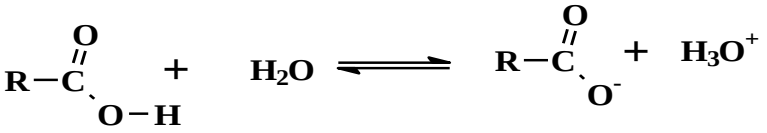
Definitions:

- **Acid:** An organic compound containing a functional group that can donate a proton (H+)
- **Base:** An organic compound that contains a functional group that can accept a H+

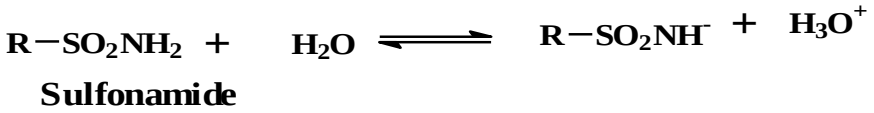
2- Recognition of acidic or basic organic functional groups

1- Common acidic organic functional groups

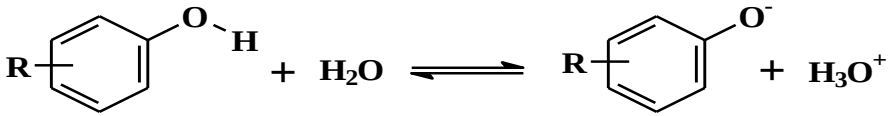
- ✓ Carboxylic acid (-COOH)
- ✓ Phenol (Ar-OH)
- ✓ Sulfonamide (R-SO2NH2)
- ✓ Imide (R-CO-NH-CO-R)
- ✓ β-Carbonyl group (-CO-CHR-CO-)



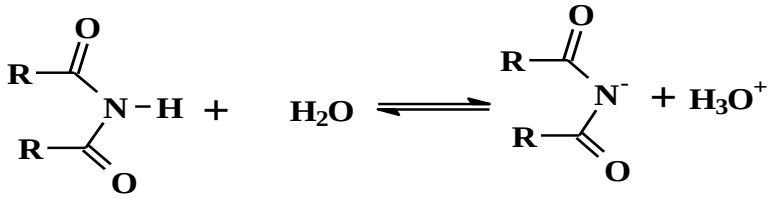
Carboxylic acid



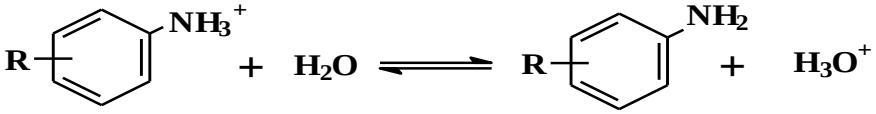
Sulfonamide



Phenol



Imide

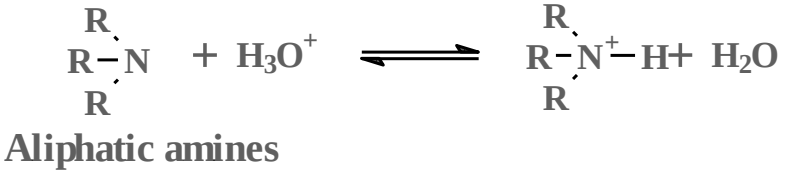


Anilinium cation

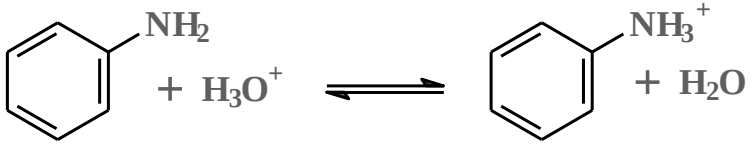
2-Recognition of acidic or basic organic functional groups(cont)

2- Common basic organic functional groups

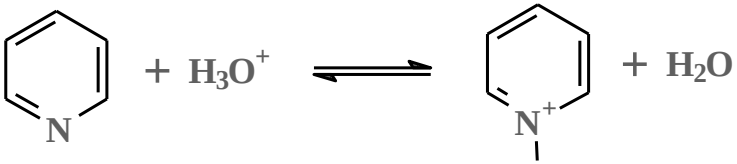
- ✓ Aliphatic 1º (R-NH2), 2º (R2NH) and 3º (R3N)-amines
- ✓ Heterocyclic amines
- ✓ Aromatic amines (Ar-NH2)



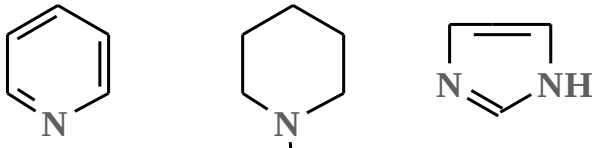
Aliphatic amines



Aromatic amines



Heteroaromatic amines



Pyridine

Piperidine

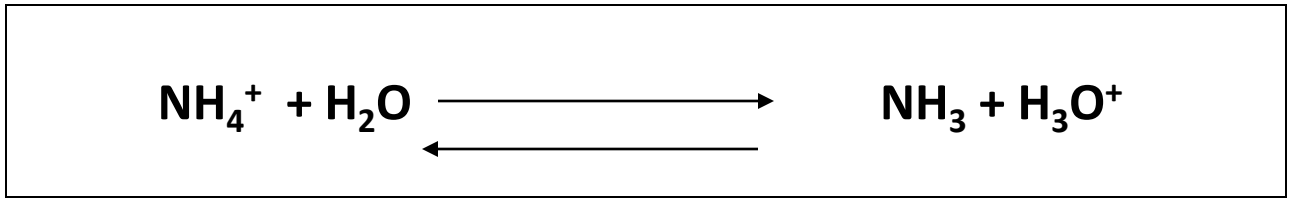
Imidazole

Estimation of the Relative Acid/Base Strength

- The ionization constant **(ka)** indicates the relative strength of the acid or base.
  - ✓ An acid with a ka of 1x10-3 is stronger acid (more ionized) than one with a ka of 1x10-5
  - ✓ A base with a ka of 1x10-7 is weaker (less ionized) than one with a ka of 1x10-9
- The negative log of the ionization constant **(pka)** also indicates the relative strength of the acid or base.
  - ✓ An acid with a pka of 5 (ka=1x10-5) is weaker (less ionized) than one with pka of 3
  - ✓ Whereas a base with a pka of 9 is stronger (more ionized) than one with a pka of 7

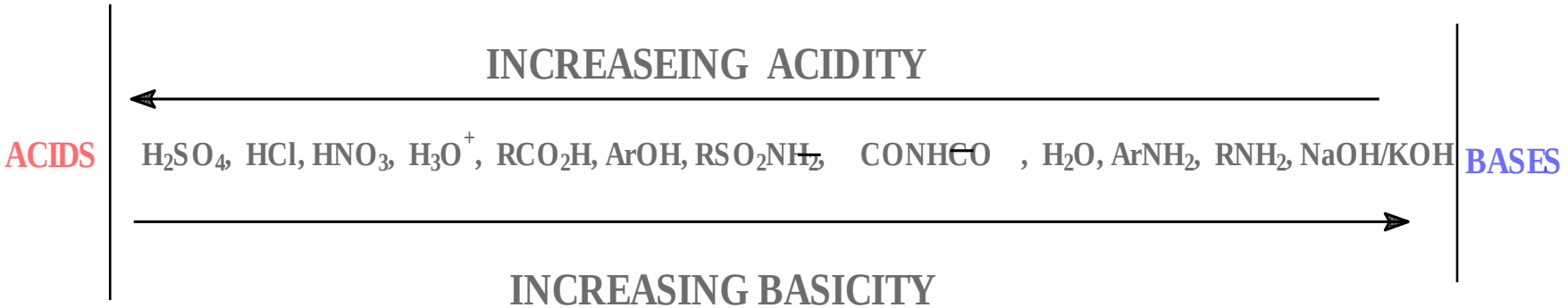
E.g. Ionization of weak acid (e.g. acetic acid, pka =4.76) is as follows:



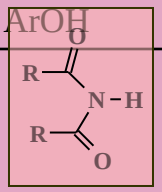


Estimation of the Relative Acid/Base Strength

The following chart is comparing acid/base strengths:



The following chart is comparing acid strengths of various functional groups

ACID	NAME	ACIDITY pKa
RSO_3H	Sulfonic acid	1
RCOOH	Carboxylic acid	4.5
$ArSO_2NHR$	Aromatic sulfonamide	6-9
ArOH	Phenol	8-11
	Imide	8-10

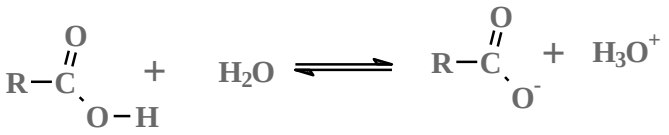
The following chart is comparing base strengths of various functional groups

ACID	NAME	Basicity pKa
RNH_2, R_2NH, R_3N	Aliphatic amines	3-4
$ArNH_2$	Aromatic amines	9-13
Pyridine, piperidine, imidazole	Heterocyclic amines	4-12

Ionization of Acidic and Basic Functional Groups

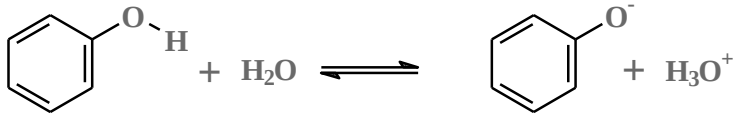
I-Acids

Carboxylic acids

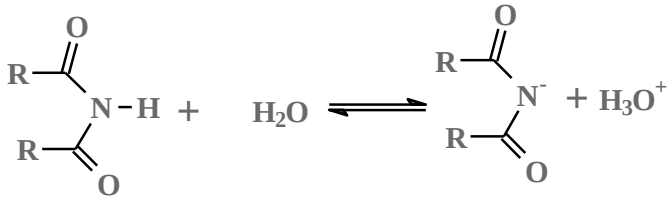


Sulfonamides





Phenols



Imides

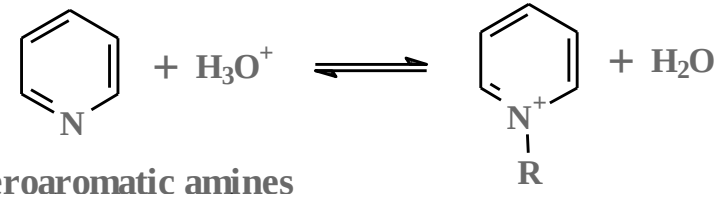
II-Bases



Aliphatic amines



Aromatic amines



Heteroaromatic amines

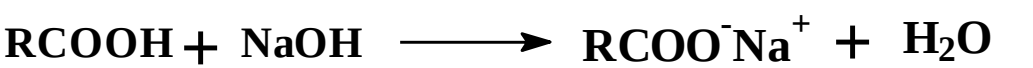
Acidic and Basic Functional Group - Salt Formation

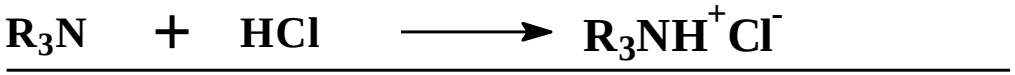
- **Salt:** is the combination of an acid and a base
- All salts are strong electrolytes (with few exceptions**:** mercuric and cadmium halides and lead acetate)
  - ✓ The salt form of the drug is more soluble than its parent molecule

Drug salts can be divided into two classes:

1) **Inorganic salts:** are made by combining drug molecules with inorganic acids and bases, such HCl, H2SO4, KOH and NaOH.
   - ✓ Inorganic salts are generally used to increase the aqueous solubility of a compound
2) **Organic salts:** are made by combining two drug molecules, one acidic and one basic.
   - ✓ The salt formed by this combination has increased lipid solubility and generally is used to make depot injections (e.g. procaine penicillin).

Sodium salt formation from carboxylic acid:





Hydrochloric salt formation from an aliphatic amine

Structurally Non-Specific and Specific Activity

❑ ***Drug activity can be classified as***:-

(a)Structurally non-specific or

(b)Structurally specific

1-Structurally non-specific activity is dependent on physical properties like solubility, partition coefficients and vapour pressure and not on the presence or absence of some chemical group.

- ✓ Substances such as alkanes, alkenes, alkynes, alcohols, amides, ethers, ketones and chlorinated hydrocarbons exhibit narcotic activity and potency of each substance is related to its partition coefficient.

Structurally Non-Specific and Specific Activity

- Structurally non-specific action results from accumulation of a drug in some vital part of a cell with lipid characteristics.

- The structurally non-specific drugs include
  - General anaesthetics,
  - Hypnotic together with a few bactericidal compounds and insecticides.

Structurally Non-Specific and Specific Activity

- Structurally specific activity is dependent upon factors such as the presence or absence of certain functional groups, intramolecular distance, and shape of the molecules.
- Activity is not easily co-related with any physical property and small changes in structure often lead to changes in activity.
- Structurally specific activity is dependent upon the interaction of the drug with a cellular receptor.

Drug-receptor Interaction

- **Receptor** is the site in the biological system where the drug exerts its characteristic effects or where the drug acts.
- Receptors have an important regulatory function in the target organ or tissue.
- Most drugs act by combining with receptor in the biological system (***specific drugs***).
  1. Cholinergic drugs interacts with acetylcholine receptors.
  2. Synthetic corticosteroids bind to the same receptor as cortisone and hydrocortisone
  3. Non-steroidal anti inflammatory drugs inhibit cyclooxygenase enzyme that will inhibit the formation of prostaglandins which will lead to inflammation symptoms.

Drug-receptor Interaction

- Non-specific drugs do not act upon receptors.

- The receptor substance is considered mostly to be a cellular constituent. Recent studies, however, indicate that the receptors are proteins or enzymes.

- The ability of a drug to get bound to a receptor is termed as the affinity of the drug for the receptor.

Drug-receptor Interaction

- The receptors are also dynamic in nature and
- have a special chemical affinity and structural requirements for the drug.
  - ✓ Thus, affinity represents kinetic constants that relate to the drug and the receptor.

- The drug elicits a pharmacological response after its interaction with the receptor.

Drug-receptor Interaction

- A given drug may act on more than one receptor differing both in function and in binding characteristics (***non-selective drugs***).
- A drug, which initiates a pharmacological action after combining with the receptor, is termed **agonist**.
- Drugs which binds to the receptors but are not capable of eliciting a pharmacological response produce receptor blockage, these compounds are termed **antagonists**.

Structural features of drugs and their pharmacological activity

- **Stereochemistry:** Space arrangement of the atoms or three-dimensional structure of the molecule.

- **Stereochemistry** plays a major role in the pharmacological properties because
  1. Any change in stereospecificity of the drug will affect its pharmacological activity
  2. The isomeric pairs have different physical properties (partition coefficient, pka, etc.) and thus differ in pharmacological activity.

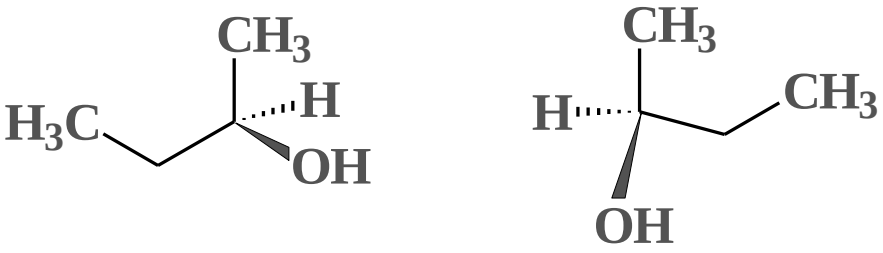
- The following steric factors influence pharmacological activity:
  - ✓ **Optical and geometric isomerism**
  - ✓ **Conformational isomerism**
  - ✓ **Isosterism and bioisosterism**

I-Optical and geometric isomerism and pharmacological activity

- ***Optical isomers*** are compounds that contain at least one chiral carbon atom **or** are compounds that differ only in their ability to rotate the polarized light.

- The (+) or dextrorotatory: isomer rotates light to the right (clockwise).
- The (-) or levorotatory: isomer rotates light to the left (counterclockwise).

I-Optical and geometric isomerism and pharmacological activity



2-Hydroxybutane enantiomers (mirror images can not superimposed)

❑**Enantiomers (optical isomers)** can have large differences in potency, receptor fit, biological activity, transport and metabolism.

- ✓ **For example, levo-phenol** has narcotic, analgesic, and antitussive properties, whereas its mirror image,
- ✓ **dextro-phenol**, has only antitussive activity.

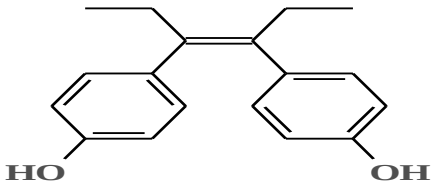
I-Optical and geometric isomerism and pharmaco-logical activity

Geometric isomerism (*cis-trans* isomerisms).

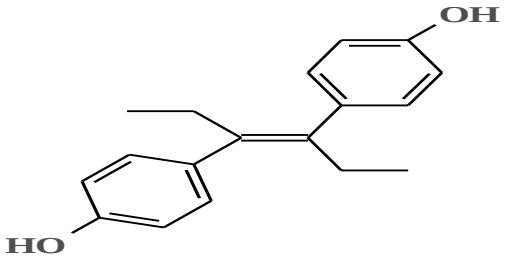
- Occur as a result of restricted rotation about a chemical bond, owing to double bonds or rigid ring system in the molecule.

- They are not mirror images and have different physicochemical properties and pharmacological activity.
  - ✓ Because different distances separate the functional groups of these isomers.

- They generally do not fit to the same receptor equally well and if these functional groups are **pharmacophores** the isomers will differ in biologic activity.
  - ✓ For example, *cis*-diethylstilbestrol has only 7% of the estrogenic activity of *trans*- diethylstilbestrol



Cis-diethylstilbestrol

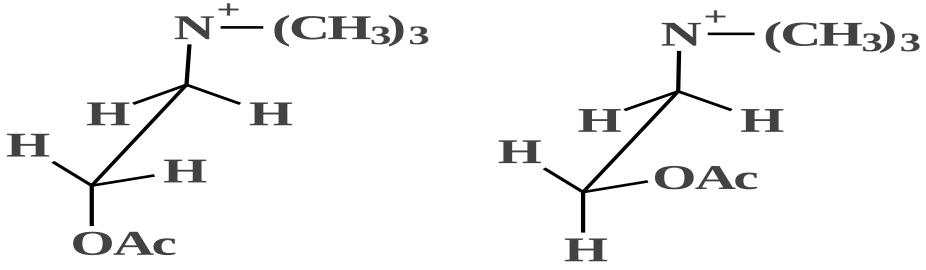


Trans-diethylstilbestrol

II- Conformational isomersim and pharmacological activity

- ***Conformational isomersim*** is the non-identical space arrangement of atoms in a molecule, resulting from rotation about one or more single bonds.

- Almost every drug can exist in more than one conformation and thus the drug might bind to more than one receptor but a specific receptor site may bind only to one of many conformations of a drug molecule.
  - ✓ For example, the *trans* conformation of **acetylcholine** binds to the **muscarinic** receptor, where as the *gauche* conformation binds to the **nicotinic** receptor.



Conformations of acetylcholine

III- Isosterism, Bioisosterism and pharmacological activity

❑***Isosterism*:** Any two ions or molecules having an identical number and arrangement of electrons

(e.g. CO and NO2; CO2(O=C=O) and N2O ($^{-}$N=N^{+}=O N= N^{+} O) ;

and N-3 and NCO- etc.).

❑**Bioisosterism** is the procedure of the synthesis of structural analogues of a *lead compound* by substitution of an atom or a group of atoms in the parent compound for another with similar electronic and steric characteristics.

- ✓ Bioisosetres are functional groups which have similar spatial and electronic character, but they retain the activity of the parent.

Bioisosterism is important in medicinal chemistry because:

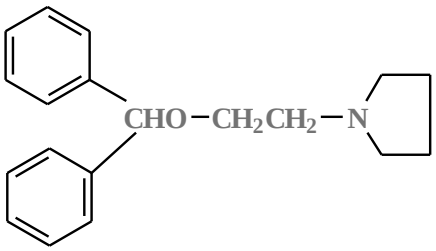
1-Maintain similar biological properties.

2-Resolved biological problems effectively (potency, side effects, separate biologic activities and duration of action)

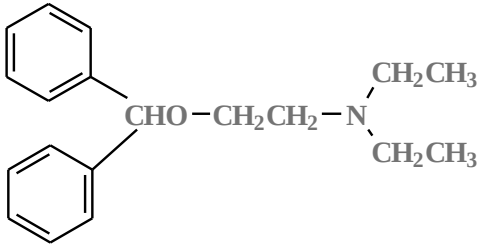
III- Isosterism and pharmacological activity

❑**Friedman** defined **bioisosterism** as- *the phenomenon by which compounds usually fit the broadest definition of isosteres and possess the same type of biological activity.*

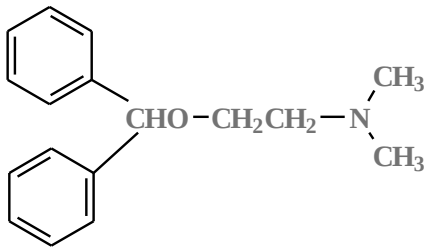
E.g. (Antihistamine; A; B and C)



A



B



C

Compound A has twice the activity of **C**, and many times greater than **B**

Classical and non-classical bioisosteres

- For the classical ones, where size equivalence is the key, the replacement should have roughly the same size.
  - ✓ The key replacements (for example, the C, O, and N replacements are seen for three of the classical isosteres: CH3-,- OH,- NH2 for univalent;
  - ✓ -CH2-, -O-, and -NH- for divalent; and -COCH2-R (ketone), -COOR (ester), and- CONHR (amide) for the carbonyl containing compounds.

- You should also be able to make isosteric replacements for the ring equivalents (single aromatic rings; single aliphatic rings, or the general tricyclic replacement)
  - ✓ For example, we could change the ester alcohol oxygen (not the carbonyl oxygen) with a CH2 (ketone), NH (amide), or S (thioester).

Drug metabolism

- Drugs are subject to attack from a range of metabolic enzymes once they inter our body
  - ✓ The role of these enzymes is to degrade or modify the foreign structure such that it can be more easily excreted

- Most drugs undergo some form of metabolic reaction resulting in structures known as metabolites
  - ✓ The metabolites may lose the activity of the original drug,
  - ✓ In some cases they may retain certain level of activity
  - ✓ Some metabolites may posses a different activity from the parent drugs, resulting in side effects or toxicity

Cont'd

- A knowledge of drug metabolism and its possible consequence can aid the medicinal chemist:-
  - ✓ In designing new drugs which do not form unacceptable metabolites
  - ✓ Possible to take advantage of drug metabolism to activate drugs in the body, this is known as prodrug strategy
- Now days, if new drug is developed; it's the requirement
  - ✓ To identify all metabolites of new drug before it can be approved
  - ✓ To determined the structure and stereochemistry of each metabolites and
  - ✓ The metabolites must be tested for biological activity

Phase I and phase II metabolism

- If the drug is polar, it will quickly excreted by kidney, however non polar drugs are not easily excreted
  - ✓ The purpose of drug metabolism is to convert such compounds in to more polar molecules that can be easily excreted
- Non specific enzyme (particularly cytochrome P450 enzymes) are able to add polar functional groups to a wide variety of drugs
- There is also other set of enzymes that can reveal masked polar functional groups which might already be present in the drug
  - ✓ Ex. There are enzymes which can demethylate a methyl ether to reveal more polar hydroxyl group

- These reactions are classed as phase I reaction and generally involves oxidation, reduction and hydrolysis
- Most of these reactions occur in the liver but some (such as hydrolysis of esters and amides can also occur in gut wall, blood plasma and other tissues)
- Some of the structure most prone to oxidation are
  - ✓N- methyl group,
  - ✓Aromatic rings,
  - ✓The terminal position of alkyl chain, and
  - ✓The list hindered position of alicyclic rings.

- Nitro, azo and carbonyl groups are prone to reduction by reductases
- Amides and esters are prone to hydrolysis by esterases
- For some drugs two or more metabolic reactions might occur resulting in different metabolites other drugs may not be metabolized at all.

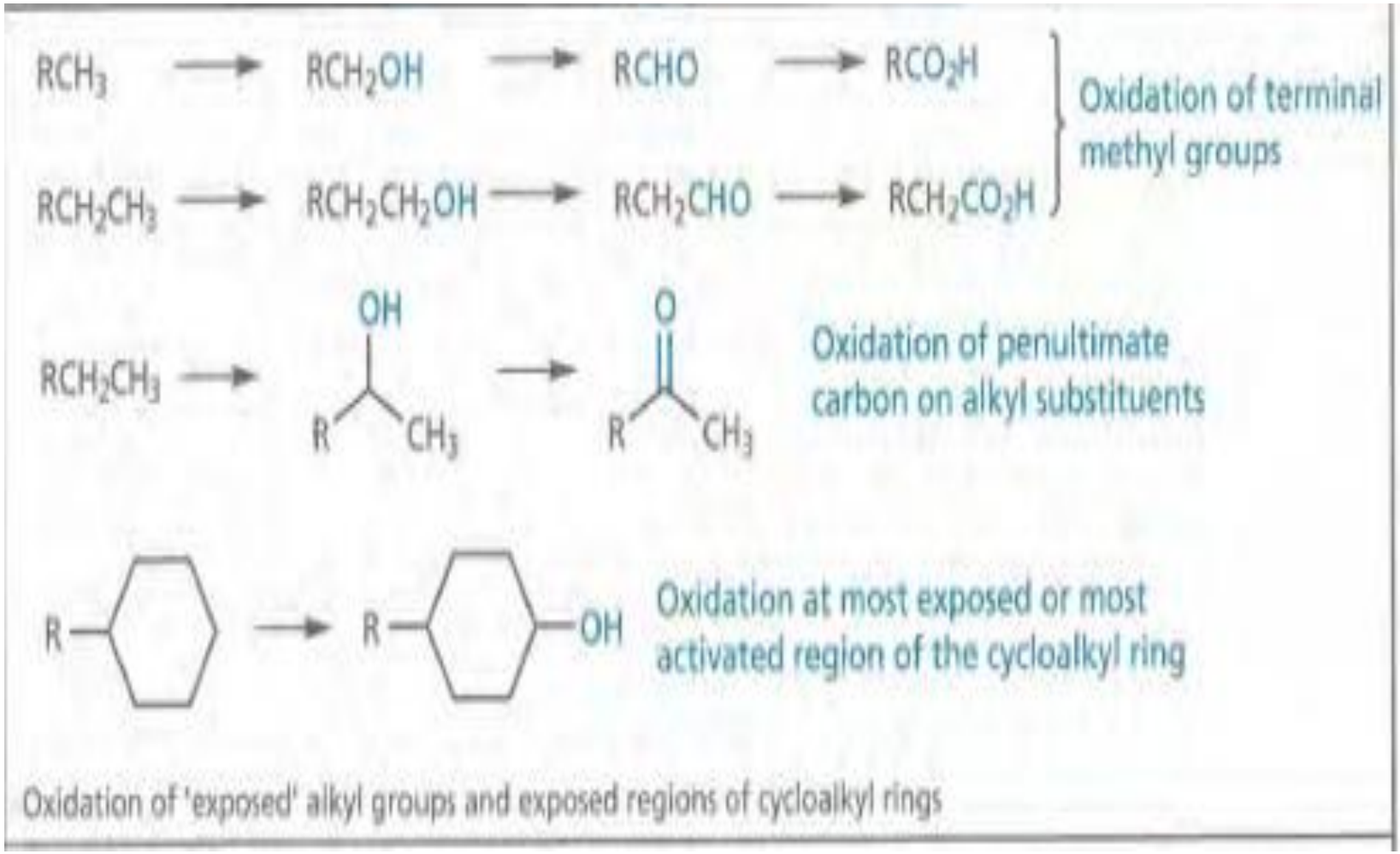
- A knowledge of metabolic that are possible for different functional groups allows the medicinal chemist to predict the likely metabolic products for any given drugs

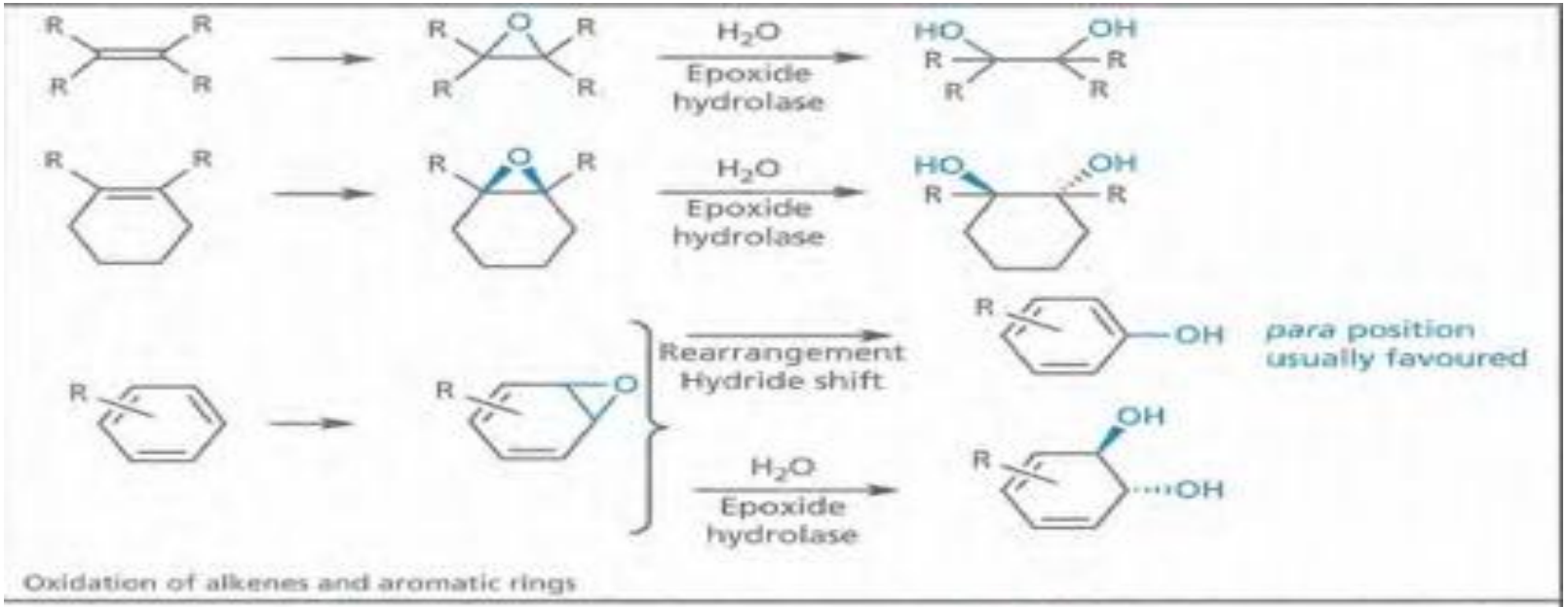
- Drug metabolism has important implication when it comes to using chiral drugs, especially if the drug is to be used as a racemate
  - One enantiomer undergo different metabolic reaction from the other
    - The enzyme involved in catalyzing metabolic reactions will often distinguish between the two enantiomers of chiral drugs
- Both enantiomers of chiral drug have to be tested separately to see what metabolites are formed
- In practice its preferable to use single enantiomer for medicine or design the drug that is asymmetric

- A serious of metabolic reaction classed as phase II reactions also occur mainly in the liver
  - ✓Most of this type of reactions are conjugation reaction
    - Attaching polar molecule to a suitable polar handle that is already present on the drug or has been introduced by phase one reaction
- The resulting conjugate has greatly increased polarity, those increasing its excretion through urine or bile
- Phase I and II reaction can be species specific, which has implication for in vivo metabolic studies

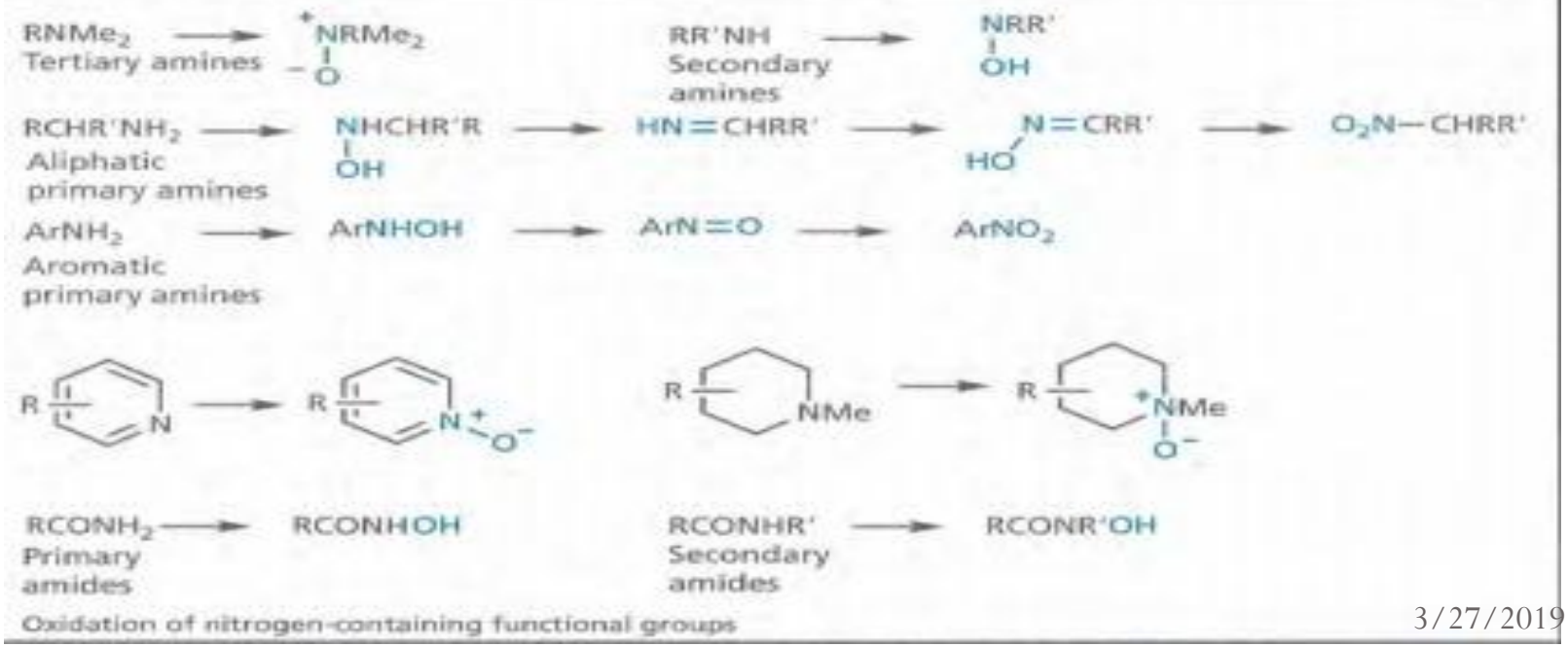
- Metabolites formed in an experimental animal may not be necessarily be those formed in humans
- Both set of reactions can be regioselective and stereoselective

Phase I transformation catalyzed by cytochrome P450 enzyme

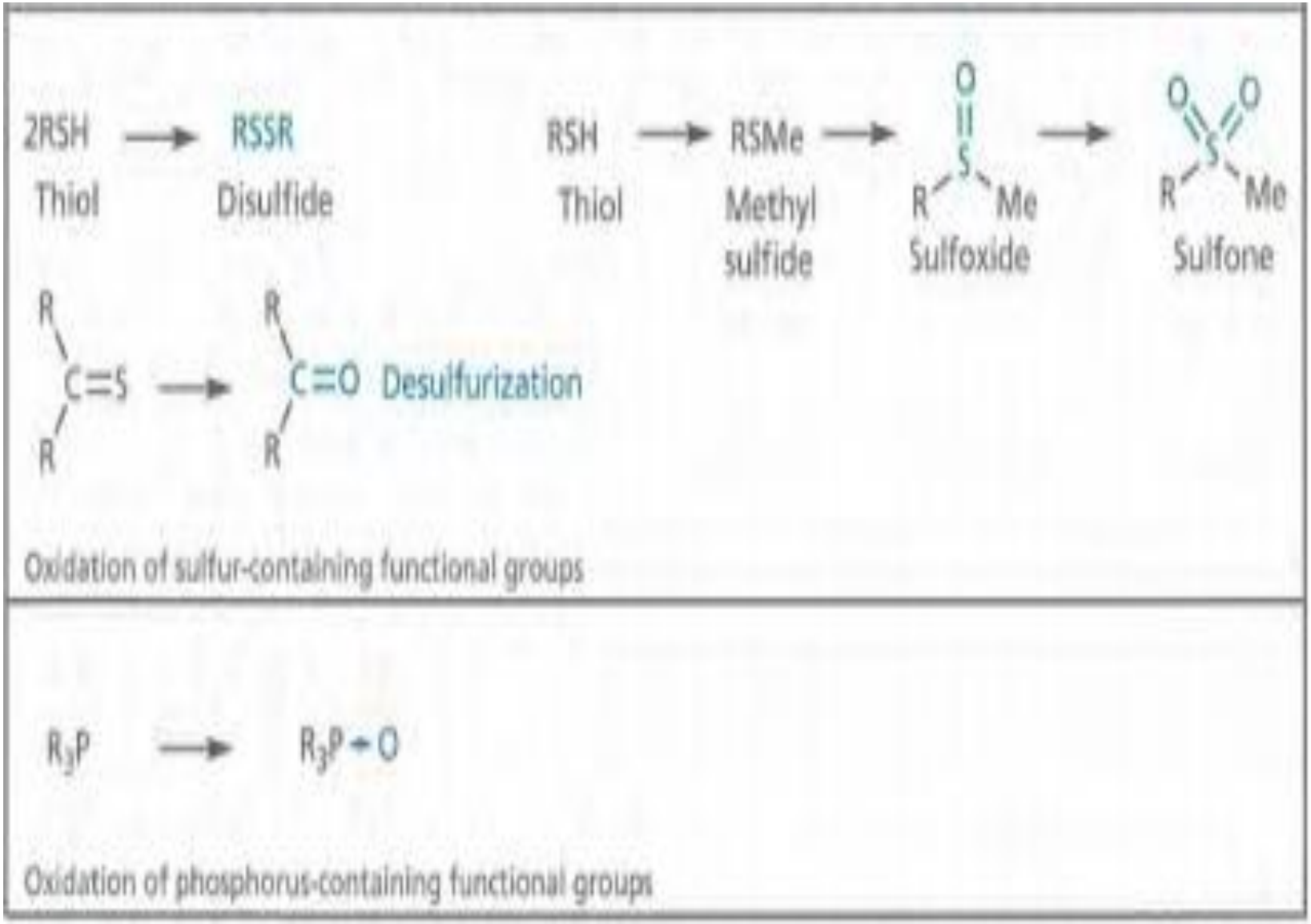




Oxidation of alkenes and aromatic rings

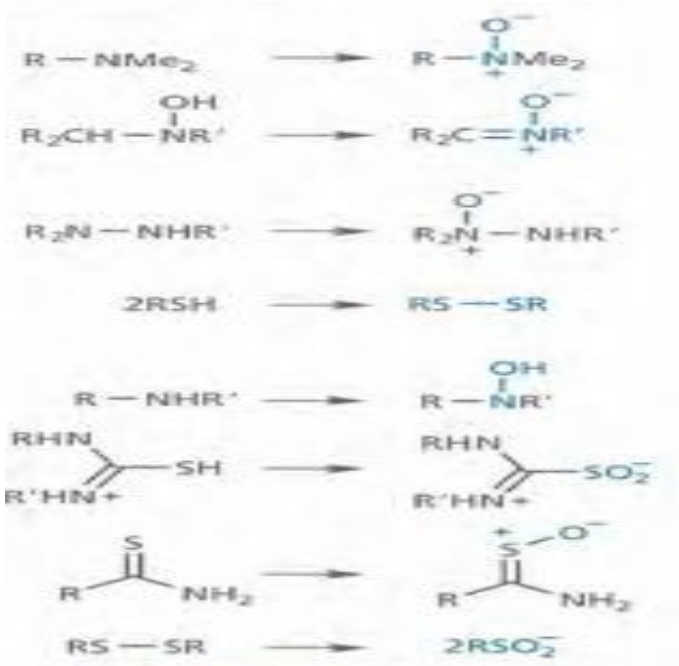


Oxidation of nitrogen-containing functional groups


2RSH
Thiol
RSSR
Disulfide
RSH
Thiol
RSMe
Methyl sulfide
Sulfoxide
Sulfone
Me
Desulfurization
Oxidation of sulfur-containing functional groups
Oxidation of phosphorus-containing functional groups

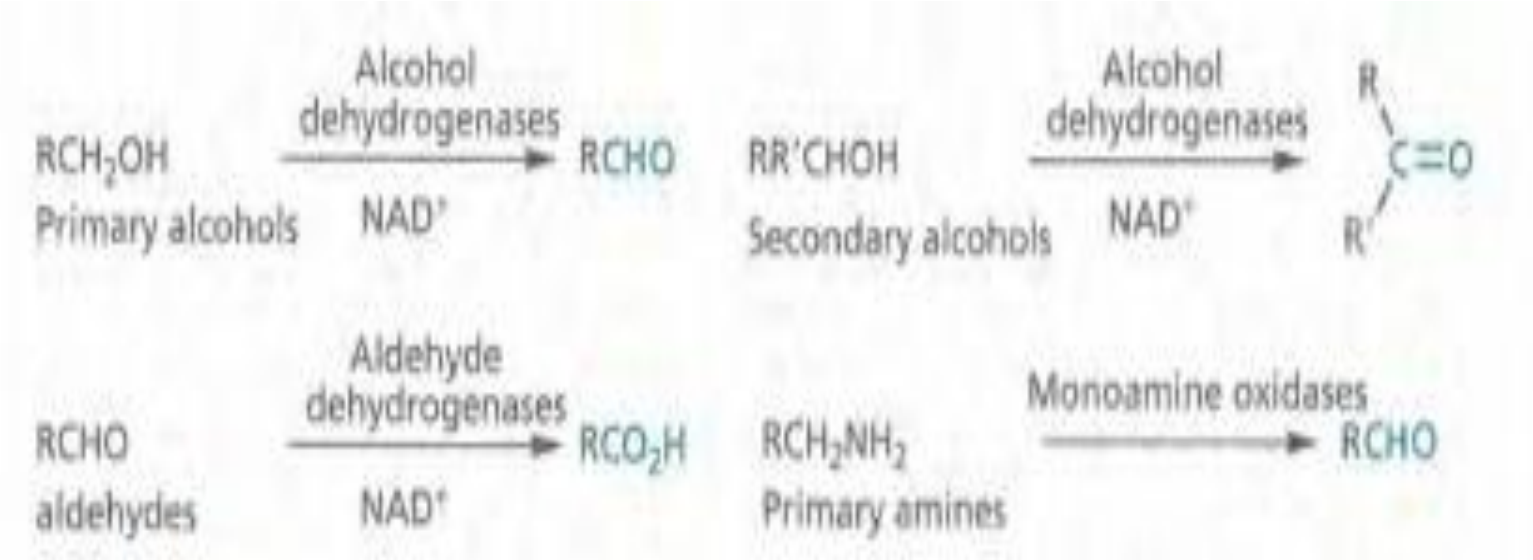
Phase I reaction catalyzed by flavin-containing monoxgenases

- Responsible for metabolic reaction involving oxidation of nucleophilic nitrogen, sulfur and phosphorus atoms rather than carbon atoms



Phase I transformations catalyzed by other enzymes

- Monoamine oxidases
- Alccohol dehydrogenase
- Aldehyde dehydrogenase



Phase II transformations

- Most of them are conjugation reaction catalyzed by transferase enzymes.
- The resulting metabolites are usually inactive but there is exception to this rule

Drug discovery, design & development; antihistamines, cimetidine

The smith Kline and French way!!

- The three steps are drug discovery, design and development

1. Drug discovery
   - ✓Identify disease target
   - ✓Identify drug target
   - ✓Identify bioassay
   - ✓Find a lead compound
   - ✓Determination of structure of the lead

2. Drug design

- Identify the SAR
- Identify the pharmacophore
- Improve target interaction
- Improve pharmacokinetic interactions

3. Drug development

- Preclinical
- Clinical studies

Drug discovery

- Choose a target disease
- Choose a drug target
- Identification of bioassay
- Find a lead compound
- Determination of the structure of the lead

- Drug discovery is a very tiresome and expensive endeavor

  - Takes about 10- 15 years
  - Cost: $800 million to $2 billion
  - Involves several departments,
    - Marketing
    - Medicinal chemistry
    - Pharmacology, pharmaceutics, pharmacognosy etc…

Search for a needle in a haystack



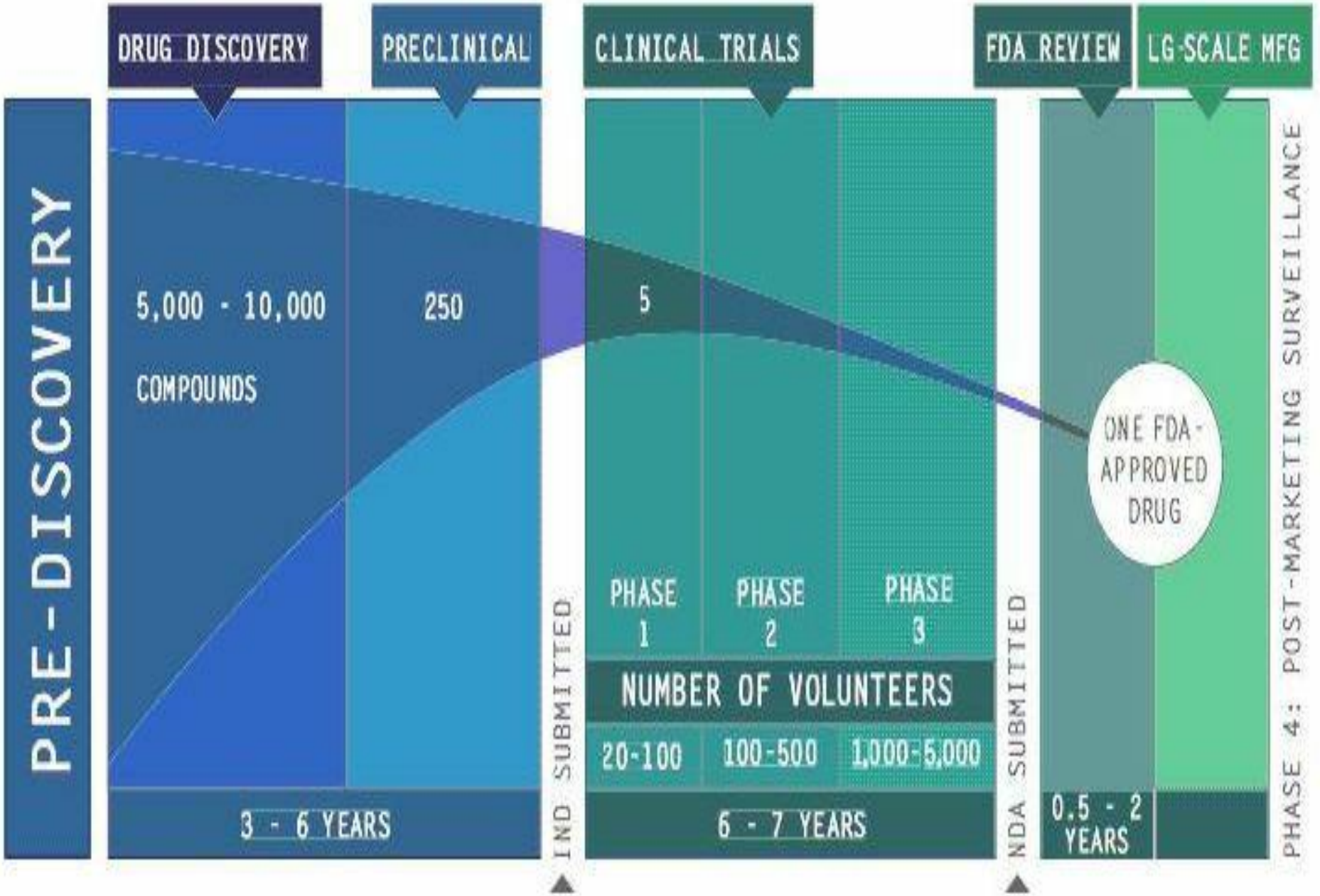
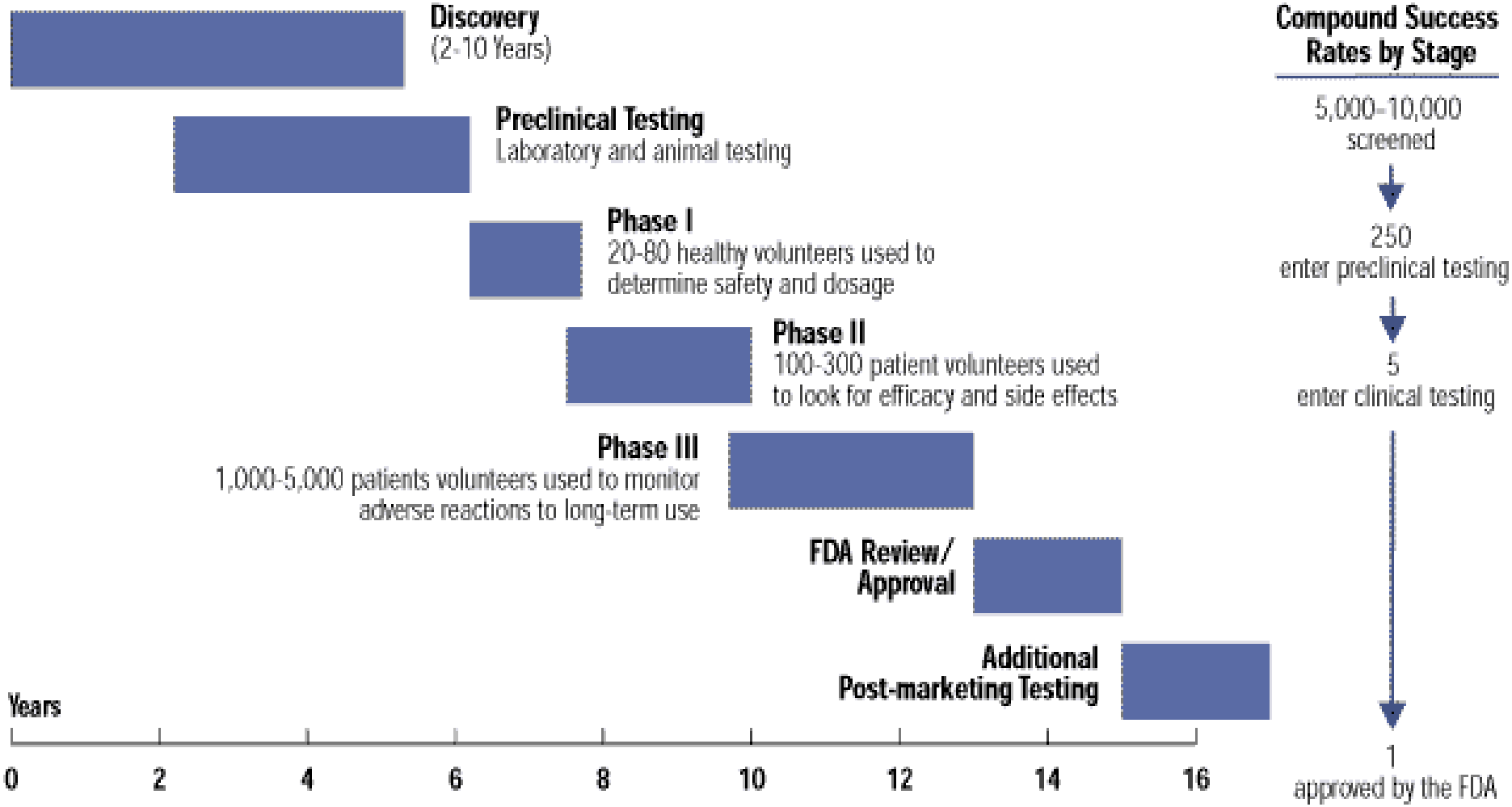

DRUG DISCOVERY
PRECLINICAL
CLINICAL TRIALS
FDA REVIEW
LG-SCALE MFG
PRE-DISCOVERY
5,000 - 10,000
COMPOUNDS
250
5
ONE FDA-APPROVED DRUG
PHASE 1
PHASE 2
PHASE 3
NUMBER OF VOLUNTEERS
20-100
100-500
1,000-5,000
IND SUBMITTED
NDA SUBMITTED
3 - 6 YEARS
6 - 7 YEARS
0.5 - 2 YEARS
PHASE 4: POST-MARKETING SURVEILLANCE

Figure 3-1

COMPOUND SUCCESS RATES BY STAGES



Source: PhRMA, based on data from Center for the Study of Drug Development, Tufts University, 1995.

"It is your *Attitude* not your *Altitude* that shows your *Aptitude*"

Wish you all the best!!